

Video Compression Homework 1 Report

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Objective

This assignment aims to explore color space transformations using a given image (`lena.png`).
The task includes:

- Converting the image from RGB to:
 - YUV color space
 - YCbCr color space
- Generating grayscale images of the individual channels:
 - $RGB \rightarrow R, G, B$
 - $YUV \rightarrow Y, U, V$
 - $YCbCr \rightarrow Y, Cb, Cr$

All transformations are done manually using the provided formulas without using any built-in color conversion functions.

Input Example: lena.png



Input Figure: lena.png

1. RGB Components

Due to OpenCV's default image reader setting being BGR, a BGR-to-RGB conversion is required for convenience before further processing.

Each color channel (R, G, B) was extracted and saved as a grayscale image.



Figure 1: Red Channel



Figure 2: Green Channel

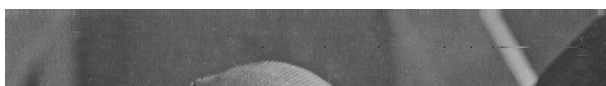




Figure 3: Blue Channel

2. YUV Transformation

The RGB to YUV transformation uses the following formulas:

$$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$

$$U = -0.169 \cdot R - 0.331 \cdot G + 0.5 \cdot B + 128$$

$$V = 0.5 \cdot R - 0.419 \cdot G - 0.081 \cdot B + 128$$

$$= 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B = -0.169 \cdot R - 0.331 \cdot G + 0.5 \cdot B + 128 = 0.5 \cdot R - 0.419 \cdot G - 0.081 \cdot B + 128$$



Figure 4: Y Channel (YUV)

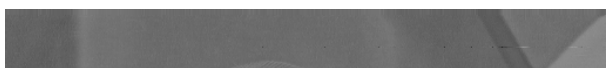




Figure 5: U Channel (YUV)

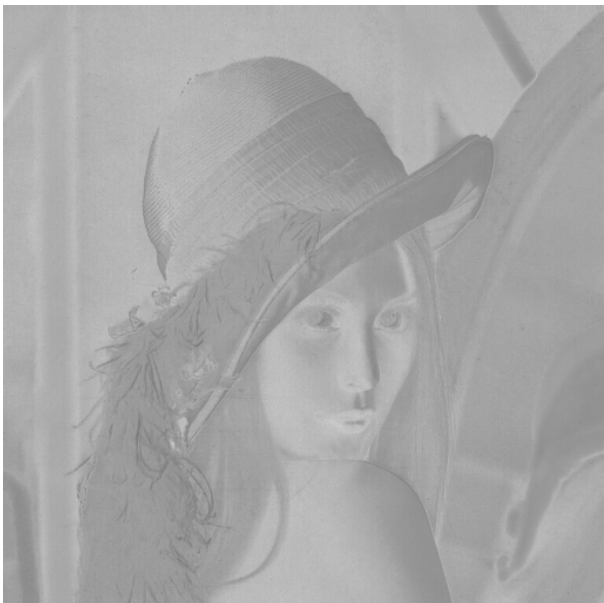


Figure 6: V Channel (YUV)

3. YCbCr Transformation

The RGB to YCbCr transformation uses the following formulas:

$$\begin{aligned}
 Y &= 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \\
 Cb &= -0.168736 \cdot R - 0.331264 \cdot G + 0.5 \cdot B + 128 \\
 Cr &= 0.5 \cdot R - 0.418688 \cdot G - 0.081312 \cdot B + 128
 \end{aligned}$$

$$\begin{aligned}
 Y &= 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \\
 Cb &= -0.168736 \cdot R - 0.331264 \cdot G + 0.5 \cdot B + 128 \\
 Cr &= 0.5 \cdot R - 0.418688 \cdot G - 0.081312 \cdot B + 128
 \end{aligned}$$





Figure 7: Y Channel (YCbCr)

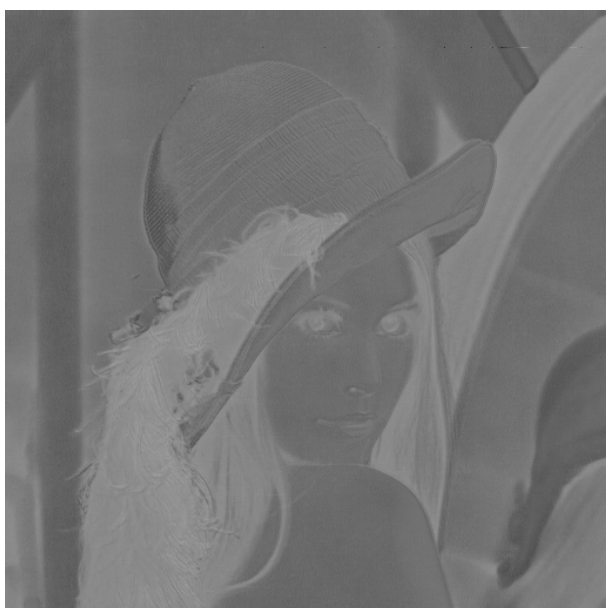


Figure 8: Cb Channel (YCbCr)

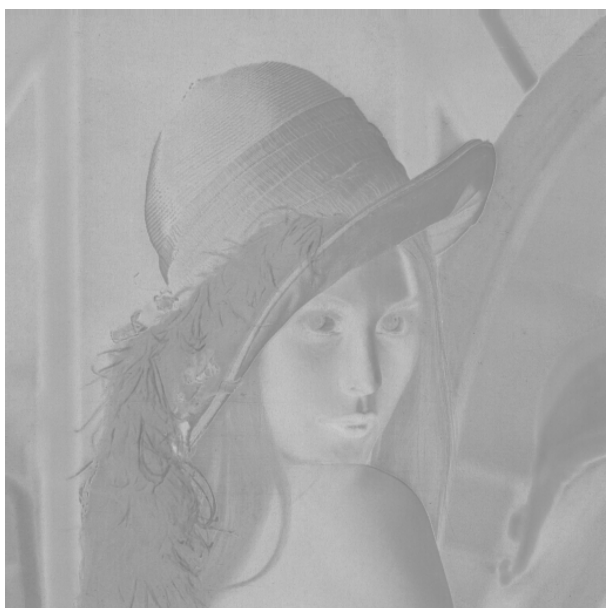


Figure 9: Cr Channel (YCbCr)

4. Implementation Notes

The project dependencies can be managed either through a `requirements.txt` file or by using `uv` (ref: [README.md](#)).

For the implementation, I implemented with "native-python" (readable) and "NumPy" (fast) version:

main.py:

- implemented in native Python with clear and readable logic.

numpy_main.py:

- implemented using NumPy arrays to achieve accelerated computation.

5. Conclusion

This assignment demonstrates manual implementation of **RGB** to **YUV** and **YCbCr** transformations and visualization of each color channel. Understanding the transformation formulas helps reinforce how image data is represented and manipulated in different color spaces.